

SALSABIL OUNI

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PROFESSIONAL EXPERIENCE

Arcadia

2025/7 - 2025/9

Machine Learning Engineer Intern

Osaka, Japan

- Developed an Arabic Text-to-Speech system using **VITS** (Variational Inference with adversarial learning), combining Conditional VAE and **GAN**-based architectures to generate natural speech with human-like prosody achieving **97%** intelligibility based on Spectrogram analysis.
- Deployed a real-time speech synthesis platform using **Python, Svelte, and Docker** with **<1.5s** latency.

PURA Solutions

2024/6 - 2024/10

Artificial Intelligence Engineer Intern

Sousse, Tunisia

- Built a medical AI system for **Thoracic X-ray** analysis and report generation using **CNN** and **Transformer** architectures, achieving **92%** validation performance.
- Developed an interactive deployment pipeline using Streamlit for model evaluation and visualization.

Systems Nakashima

2024/2 - 2024/5

Machine Learning and Full Stack Research Engineer Intern

Tokyo, Japan

- Developed **Genki Walk**, an AI rehabilitation system used by **36+ patients**, combining **MediaPipe** pose estimation and **Transformer** models for gait analysis.
- Built and deployed a healthcare platform using **Django, Vue.js, and AWS**, achieving **91% gait analysis accuracy**.

Systems Nakashima

2023/8 - 2023/10

Artificial Intelligence and Full Stack Engineer Intern

Tokyo, Japan

- Developed **Aintone**, an AI-powered sales forecasting platform using **PyTorch** based machine learning **models** for predictive analytics and built an end-to-end application integrating **Django** backend and **Vue.js** frontend.
- Presented the project at **Tokyo Big Sight 2023**, attracting **400+** attendees and receiving positive feedback from industry professionals.

Envast

06/2023 – 08/2023

Machine Learning Engineer Intern

Sousse, Tunisia

- Developed an NLP chatbot for automated CV and cover letter generation using **TensorFlow, SpaCy, and Django** and implemented text processing pipelines for personalized document generation.

EDUCATION

Polytechnic School of Sousse

2024 - Present

Engineering Diploma in Artificial Intelligence and Data Science

Faculty of Medicine Ibn El Jazzar of Sousse

2024 - Present

Master's Degree in Molecular Medicine : AI Applied to Genetics, Immunology and Microbiology

Higher Institute of Applied Science and Technology of Sousse

2021 - 2024

Bachelor of Computer Science and Software Engineering, with High Honors

ADDITIONAL INFORMATION

Technical Skills: Python | PyTorch | TensorFlow | Scikit-learn | NumPy | Pandas | Transformers | Hugging Face | LLMs | Generative AI | RAG | NLP | Computer Vision | CNNs | RNNs | LSTMs | GANs | Fine-tuning | LoRA | FastAPI | REST APIs | SQL | PostgreSQL | Apache Spark | Apache Kafka | MLflow | Docker | Kubernetes | CI/CD | Git | GitHub | AWS | Azure | Linux | C/C++ | Java | JavaScript | Django | Computational Bioinformatics

Languages: English | French | Arabic | Japanese | German

Certifications: NVIDIA Fundamentals of Deep Learning | NVIDIA Introduction to Transformer-Based Natural Language Processing | NVIDIA Generative AI with Diffusion Models | Microsoft Azure Data Fundamentals

Achievements: AIESEC Waseda Japan Global Talent 2023 selected out of 22 countries globally | AI Project Exhibitor at Tokyo Big Sight 2023 | Winner of 16 Hackathons.

Projects: Protein Secondary Structure Prediction | Yolo Queue Intelligence | 3D Protein Viewer | DNA Sequence Classification | Protein Thermostability Prediction | Breast Cancer Detection in Women.

Research Papers In Progress: Assessing the Validity of 3D Structural Patches for the Classification of Point Mutations | ML-Based Gait Analysis for TKA Rehabilitation | NLP and ML approaches to the Endometriosis-Fibromyalgia Link

Volunteering: TEDx Marketing Manager, Hadrumet (2025) | Global Ambassador, IEEE Day Region 8 (2022)